

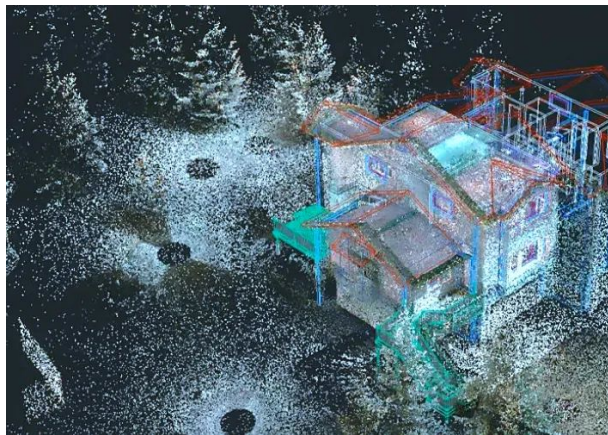
Real-time GS Mapping on ARIA

Dong Hwi Kim, Chaehyeon Song

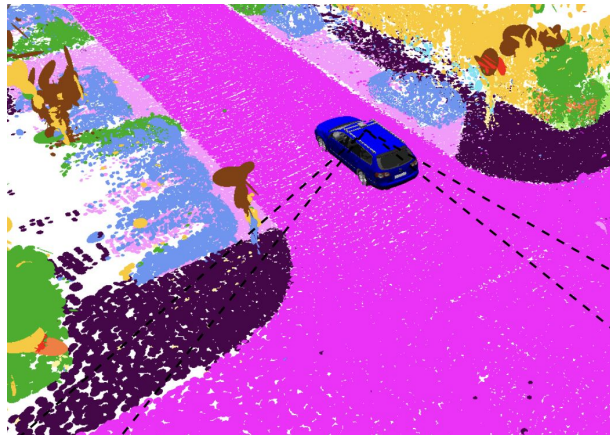
What is a mapping?

Reconstruct a N-dimensional space in a specific representation

Point Cloud



Surfel

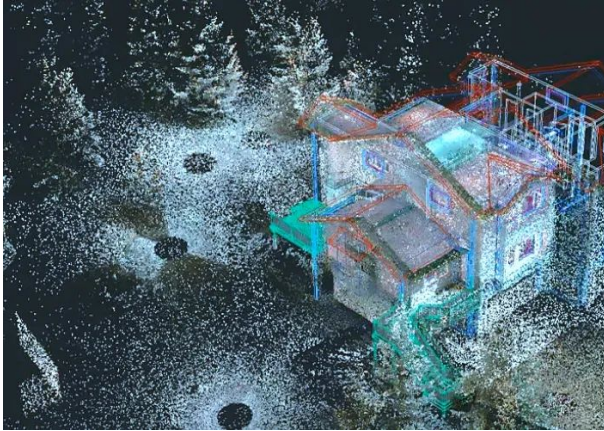


Mesh



What is Gaussian Splatting

What happens if points have volume?



Advantages of GS

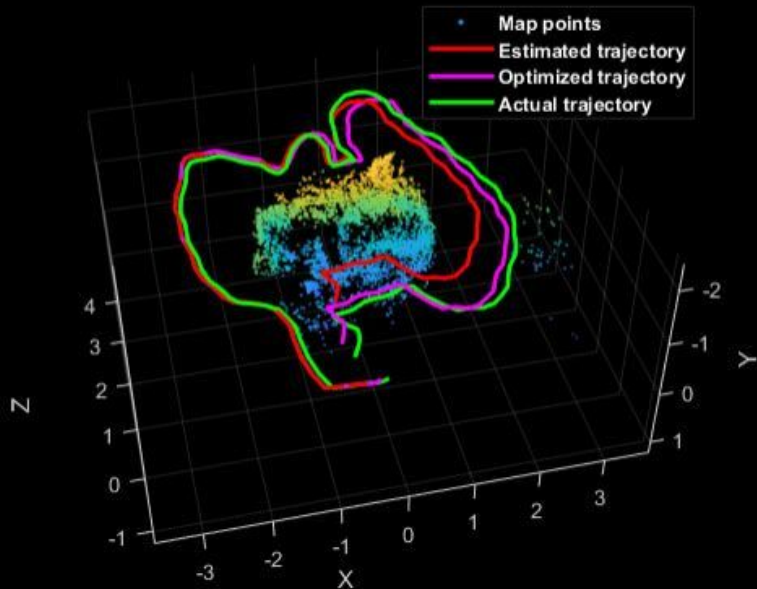
Photo realism + Fast rendering



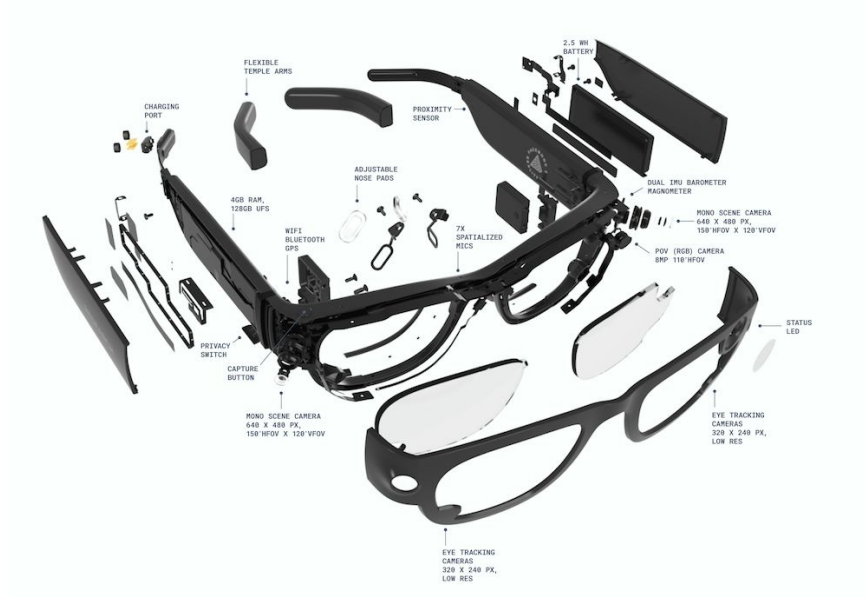
1. Kerbl, Bernhard, et al. "3D Gaussian splatting for real-time radiance field rendering." *ACM Trans. Graph.* 42.4 (2023): 139-1.

Problem: How to get image poses?

Trajectory estimation: Odometry + Localization



What is ARIA



Simultaneous localization and Mapping

SLAM = Localization (state estimation) + Mapping + global consistency

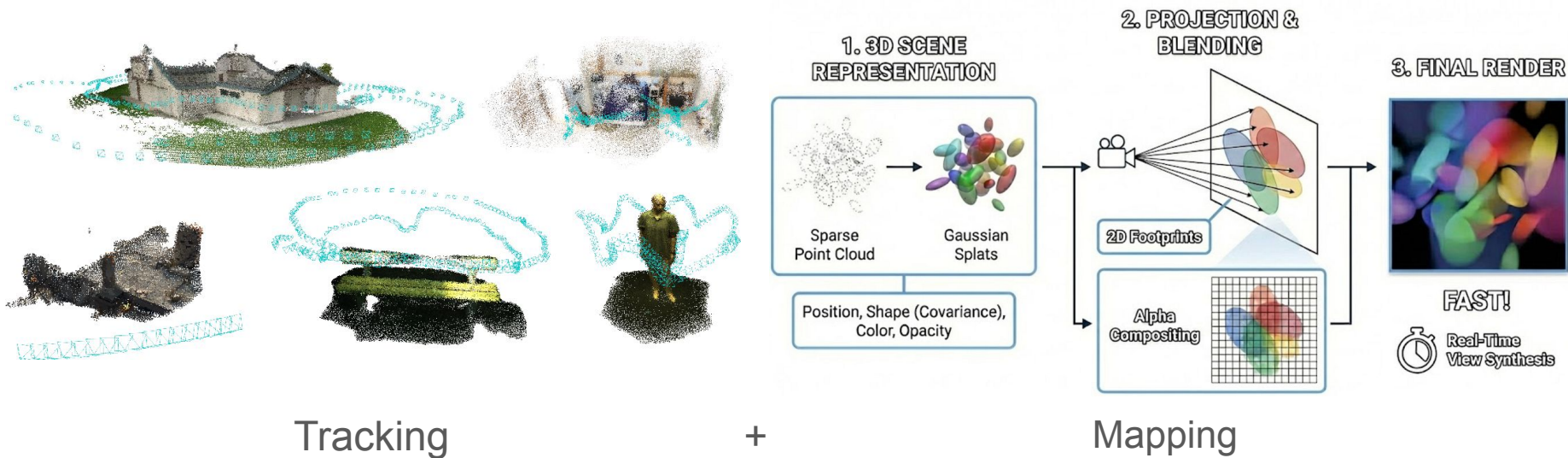
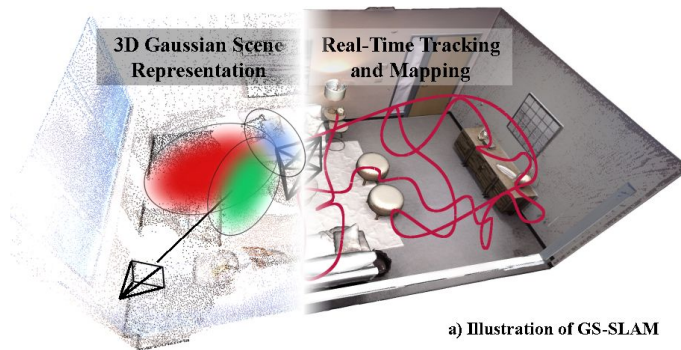


Image credits: Teed, Zachary, and Jia Deng. "DROID-SLAM: Deep visual SLAM for monocular, stereo, and RGB-D cameras." *Advances in Neural Information Processing Systems* 34 (2021): 16558–16569; Kerbl, Bernhard, et al. "3D Gaussian splatting for real-time radiance field rendering." *ACM Transactions on Graphics* 42.4 (2023): 139:1–139:14.

Online 3DGS-SLAM

Fast loop: localize every incoming frame

Mapping loop: add, refine, and remove Gaussians on selected views

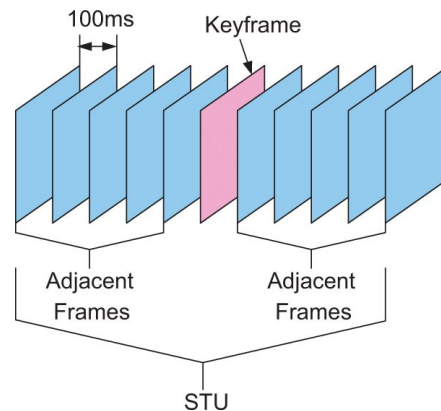
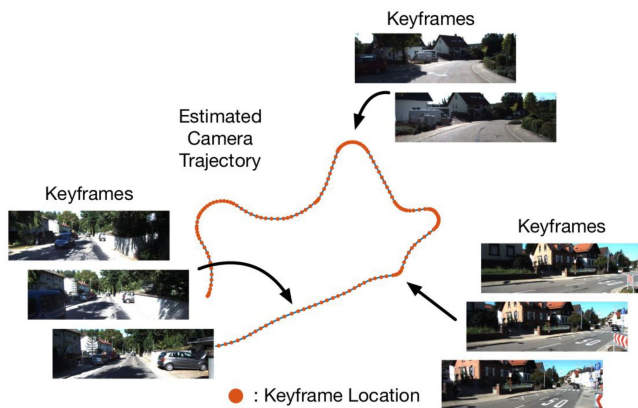


1. Yan, Yanjun, et al. "GS-SLAM: Dense Visual SLAM with 3D Gaussian Splatting." Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2024.
2. Keetha, Nikhil, et al. "SplaTAM: Splat, Track & Map 3D Gaussians for Dense RGB-D SLAM." Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2024.

Limitations of Existing Online 3DGS-SLAM Pipelines

1. A Tracking Keyframe Is Not Automatically a Good Mapping View

- Mapping needs coverage, novelty, and repeated photometric supervision
- Inter-keyframe RGB frames can still improve an existing Gaussian map

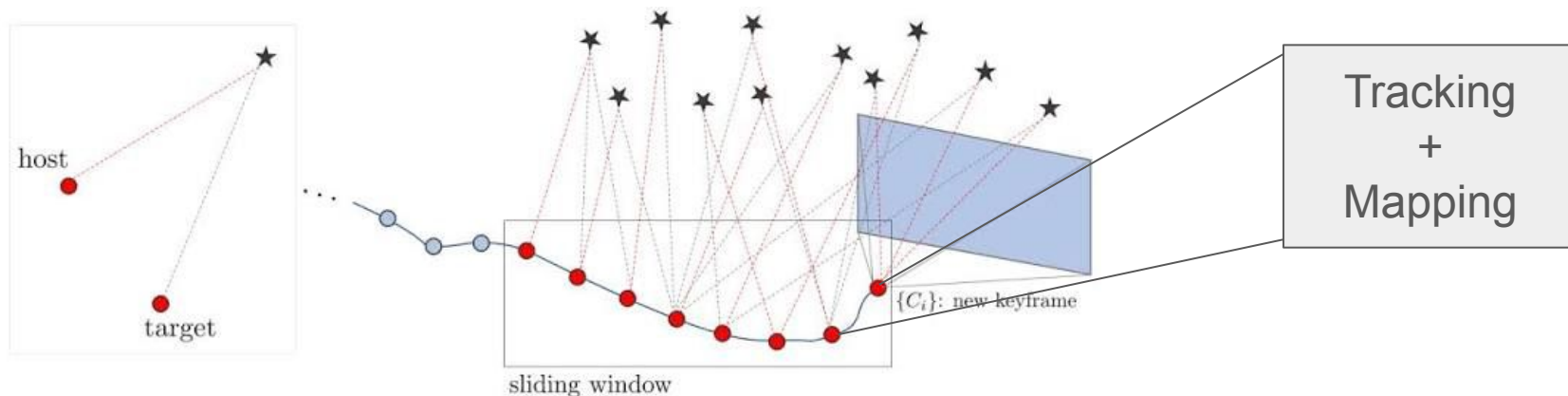


1. Dias, Nigel Joseph Bandeira, Gustavo Teodoro Laureano, and Ronaldo Martins da Costa. "Keyframe Selection for Visual Localization and Mapping Tasks: A Systematic Literature Review." *Robotics* 12.3 (2023): 88.
2. Yin, Hao, Wen Hui, Hongzhi Li, Chuang Lin, and Wenwu Zhu. "A Novel Large-Scale Digital Forensics Service Platform for Internet Videos." *IEEE Transactions on Multimedia* 14.1 (2012): 178–186.

Limitations of Existing Online 3DGS-SLAM Pipelines

2. A Fixed Mapping Rate Cannot Respect a Variable GPU Budget

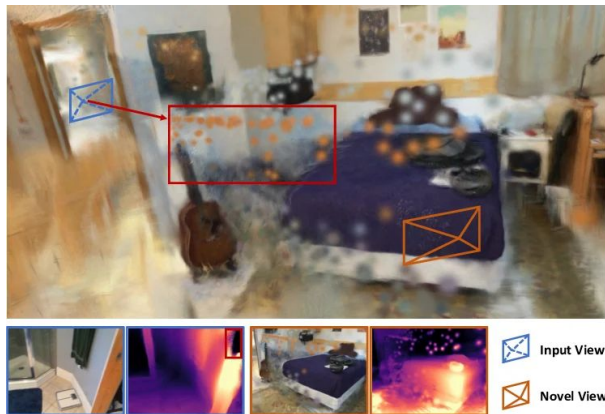
- Fixed FPS ignores the GPU time actually available after tracking
- Real time requires: tracking + preprocessing + mapping $\leq$ input time



Limitations of Existing Online 3DGS-SLAM Pipelines

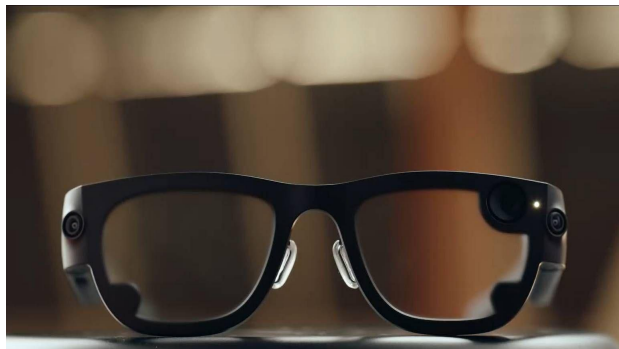
3. Floaters Expose What Rendered Error Does Not See

- RGB loss: A floater can fit pixels; depth-order swaps make center gradients discontinuous.
- Free-space correction: Moving it is indirect and risks surfaces; opacity directly suppresses its rendered contribution.



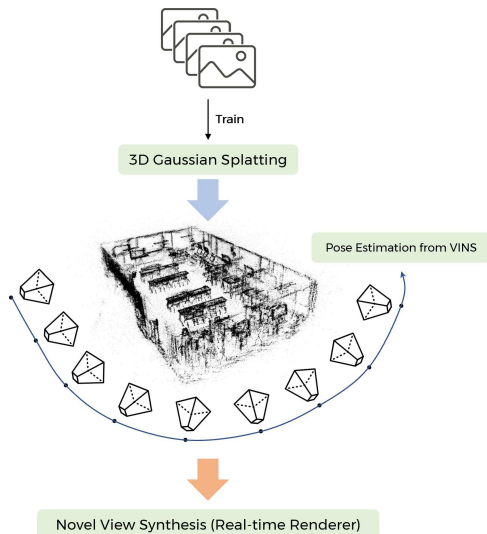
1. Wang, Yunsong, Tianxin Huang, Hanlin Chen, and Gim Hee Lee. "FreeSplat++: Generalizable 3D Gaussian Splatting for Efficient Indoor Scene Reconstruction." arXiv preprint arXiv:2503.22986 (2025).

Why This Is a Mechanical Systems Problem



State estimation under sensor uncertainty

Image credit: Meta, Project Aria Gen 2 smart glasses, via Tom's Guide (2025).



Spatial reliability under physical constraints

Image credit: lovely_yoshino, “认识3D Gaussian Splatting以及如何和ROS结合完成实时建图,” CSDN Blog, July 25, 2024.



Real-time allocation of limited sensing and compute resources

Applications: inspection, digital twins, AR, and robot interaction